# Type MS · Interpret a raised or lowered blood cell count

## Answer 1 of 4 · Question 5 part a.ii · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 5 (a) | (i) | • 13 750 ÷ 7500;<br>• 1:1.83; | Allow 1:1.8/1:2/ 1:1.8333...<br>allow if they do it the other way round ie 7500 ÷ 13750 | 2 |
| | (ii) | • person A has an infection/disease;<br>• Immune response/more white blood cells needed, to produce antibodies/phagocytosis/destroy pathogen/virus/bacteria; | ignore sick<br>ignore fight infection<br>allow alternative correct explanation | 2 |
| | (iii) | Any two from:<br>• excess bleeding;<br>• poor/no blood clotting;<br>• poor/slow wound healing/more time for pathogens to enter wound; | | 2 |
| (b) | (i) | • tiredness/fatigue/high heart rate/faster breathing;<br>• less oxygen carried/transported (to body cells);<br>• less aerobic respiration;<br>• less energy; | | 4 |
| | (ii) | • HA is dominant;<br>• parents are heterozygous;<br>• HA expressed in heterozygous form;<br>• so only needs one allele; | ignore ref to allele/gene for mpt 1 | 4 |

**Total Marks 14**

## Type MS · Interpret a raised or lowered blood cell count

### Answer 1 of 4 · Question 5 part a.ii · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Interpret a raised or lowered blood cell count

### Answer 2 of 4 · Question 1 part c.ii · 2 marks

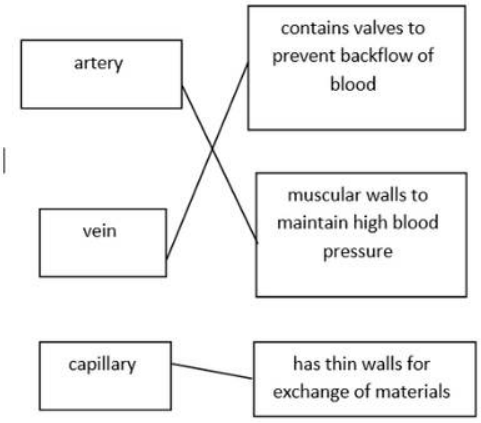
2020 NOV · Paper 01 (regional R) · find part c.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

## Type MS · Interpret a raised or lowered blood cell count

### Answer 3 of 4 · Question 2 part a.iii · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | X = plasma;<br>Y = red blood cell;<br>Z = white blood cell/phagocyte; | Accept named phagocyte | (3) |
| (a) (ii) | • transports oxygen;<br>• to body/ cells/tissues/organs; | | (2) |
| (a) (iii) | the patient has an infection/leukaemia/allergy/ blood cancer, inflammation; | Accept pathogens in body | (1) |
| (b) |  | Reject more than one line from each blood vessel<br>3 correct=2marks<br>1 or 2 correct= 1mark | (2) |
| (c) (i) | left ventricle; | | (1) |
| (c) (ii) | • into right ventricle;<br>• through pulmonary artery;<br>• to the lungs;<br>• (from the lungs) into the pulmonary vein; | | (4) |

**Total for Question 2 = 13 marks**

## Type MS · Interpret a raised or lowered blood cell count

### Answer 4 of 4 · Question 5 part a.iii · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 5 (a) | (i) | • 13 750 ÷ 7500;<br>• 1:1.83; | Allow 1:1.8/1:2/ 1:1.8333...<br>allow if they do it the other way round ie 7500 ÷ 13750 | 2 |
| | (ii) | • person A has an infection/disease;<br>• Immune response/more white blood cells needed, to produce antibodies/phagocytosis/destroy pathogen/virus/bacteria; | ignore sick<br>ignore fight infection<br>allow alternative correct explanation | 2 |
| | (iii) | Any two from:<br>• excess bleeding;<br>• poor/no blood clotting;<br>• poor/slow wound healing/more time for pathogens to enter wound; | | 2 |
| (b) | (i) | • tiredness/fatigue/high heart rate/faster breathing;<br>• less oxygen carried/transported (to body cells);<br>• less aerobic respiration;<br>• less energy; | | 4 |
| | (ii) | • HA is dominant;<br>• parents are heterozygous;<br>• HA expressed in heterozygous form;<br>• so only needs one allele; | ignore ref to allele/gene for mpt 1 | 4 |

**Total Marks 14**

## Type MS · Interpret a raised or lowered blood cell count

### Answer 4 of 4 · Question 5 part a.iii · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |